

Haptics for Accessible Graphics in CS1

William Bares
Jordan Wirfs-Brock
baresw@whitman.edu
wirfsbrj@whitman.edu
Whitman College
Walla Walla, Washington, USA

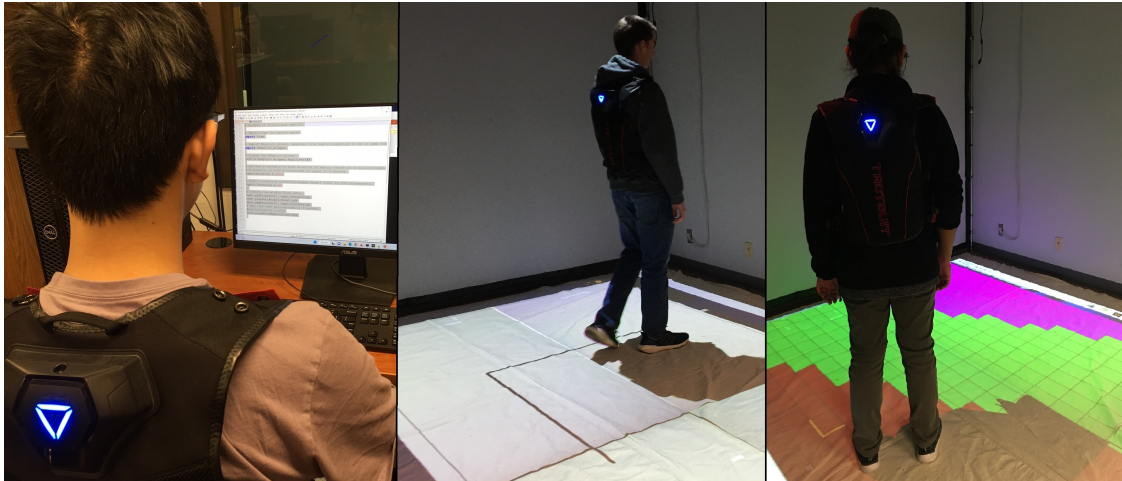


Figure 1: Turtle programming (left), Turtle navigation (middle), and haptic data exploration (right)

ABSTRACT

Tactile or haptic presentations of computer graphics visualizations can offer more inclusive experiences for persons who are blind or have low vision. However, teaching accessibility with haptics is challenging due to a lack of ready-to-use instructional resources. This Engaging Education Technique and Assignment (EETA) introduces accessibility with haptics and Turtle Graphics after only a few weeks into CS1. The sample assignments feature open-ended exploration to meaningfully map graphical attributes to haptic feedback that could benefit blind or low vision users. Supplementary materials include lesson guides and Python samples.

CCS CONCEPTS

• **Human-centered computing** → Haptic devices; **Visualization techniques**; **Accessibility technologies**.

KEYWORDS

accessibility, CS1, data physicalization, data visualization, haptics

ACM Reference Format:

William Bares and Jordan Wirfs-Brock. 2023. Haptics for Accessible Graphics in CS1. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '23 Educator's Forum)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587424.3595584>

1 OVERVIEW

Worldwide, an estimated 43.3 million persons were blind in 2020, and another 295 million had moderate to severe visual impairment. Given these statistics, accessibility should be part of computer graphics education. However, a systematic analysis of computing education research on teaching accessibility found that there were no consistent guidelines for topics, learning outcomes, teaching methods, or methods to assess student learning [Baker et al. 2020]. Integrating accessibility into computer graphics affords an opportunity to communicate information in different modalities.

Expressing visual information in tactile modalities enables people to feel physical sensations to interpret information such as Braille characters. Vibrotactile feedback devices use small motors to cause a moving part to vibrate as it contacts the wearer's skin. For example, the HapticColor wristband helps colorblind wearers interpret colors using vibrations [Carcedo et al. 2016].

This EETA introduces accessible visualization using a consumer-level bHaptics haptic vest that creates vibratory feedback of varying intensity at 16 or 40 haptic dots on the front or back of the vest. For example, a haptic vest can help a blind person navigate by vibrating

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Educator's Forum, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0146-7/23/08.

<https://doi.org/10.1145/3587424.3595584>

one side of the vest to signal the direction to turn or varying the intensity to communicate how far to walk. Haptic feedback can enable a blind person to explore a visualization by mapping colors to different areas of the vest or numeric scales to vibration intensity.

The bHaptics vest provides a minimalist Python interface for selecting dot(s) by their unique integer indices, vibration intensity between 0-100, and vibration duration in milliseconds. We provide a wrapper class to offer a Turtle-like notation to hide the list and dictionary arguments used by the bHaptics interface.

Our CS 1 class introduced Turtle Graphics [Caspersen and Christensen 2000] by the third week of the semester. We designed an optional 50-minute lesson comprising a five-minute introduction to haptics and examples of haptic aids for the blind, a pair-programming activity to understand and apply the haptic function calls, and a pair-programming challenge to modify one of three given Python programs to add haptic feedback.

1.1 Navigation Aid

Have student pairs design a way to communicate pedestrian navigation turn and distance directions. Evaluate the designs by how accurately the wearer translates haptics to navigation moves (Figure 1 middle). This Python Turtle listing vibrates dots on the right side of the vest to signal a right turn.

```
vest = bhaptics_wrapper.HapticVest()
hapticIntensity = 50
hapticDuration = 250
for i in range(4):
    Alice.forward(300)
    # Dots 3 and 11 on vest right side.
    vest.addFrontDot(3, hapticIntensity)
    vest.addFrontDot(11, hapticIntensity)
    vest.playDots(hapticDuration)
    Alice.right(90)
```

1.2 Map Data to Vibrations

Challenge students to propose a way to map a data visualization of a line graph, bar chart, or multidimensional dataset to directional vibrations and/or intensity. Evaluate the designs by how well classmates are able to correctly interpret the data.

1.3 Haptic Data Walkthrough

Invite pairs of students to do a Wizard of Oz simulation of a data walk through by having one student wear the haptic vest while an image of the dataset is projected onto a wall or floor. Their partner uses a mouse or keyboard to select the data grid cell that corresponds to the position of the haptic vest wearer as corresponding haptic feedback is generated (Figure 1 right).

2 METADATA AND CONCLUSIONS

Table 1 summarizes the metadata for this EETA.

The lesson using the navigation prompt was tested in an optional activity outside of the CS1 class. The six participants evaluated the activity on a 1-7 scale to respond to the following questions:

- (1) I found it easy to understand the tutorial (average 6.83).

Table 1: EETA metadata in tabular format

Summary	What is it? What do students learn?
Learning outcomes	Apply haptics to communicate data.
Classification(s)	Graphics and Interfaces, Visualization
Audience	High school and undergraduates in CS1
Dependencies	Access to a haptic feedback device
Prerequisites	Objects and Turtle or Zelle Graphics
Strengths	Motivation to create accessible graphics
Weaknesses	Limited number of haptic devices
Variants	Add audio cues or 3D printed forms
Assessment	Testers can interpret haptic cues

- (2) I was comfortable using the Python programming skills that I learned in my CS1 class (average 6.83).
- (3) I enjoyed the tutorial (average 6.67).
- (4) I learned how to create an interactive experience that would benefit a person having a disability (average 7.0).

Future studies will use multiple less-expensive haptic devices so that an entire class can participate together. Best practices for how to use haptics are still emerging. Thus, we see an opportunity for instructors to involve students in open-ended activities to explore a new design space [Bae et al. 2022] for using haptic feedback to communicate data. Instructional prompts might pose open-ended questions including: *What types of data are most appropriate for haptic representation? How might various strategies for encoding data via haptic feedback—spatially, temporally, using intensity, etc.—emphasize different aspects of data? What new environmental and social contexts for encountering data does a haptic vest enable?* Incorporating haptic vests into computer science classrooms has the potential to encourage students to think beyond how data is *represented* and to consider how data is *experienced*.

3 MATERIALS

Required materials include a bHaptics vest, a Windows PC with Bluetooth, bHaptics Player, Python, and Python websocket client. Use an Arduino or Raspberry Pi for a less-expensive haptic device.

REFERENCES

- S Sandra Bae, Clement Zheng, Mary Etta West, Ellen Yi-Luen Do, Samuel Huron, and Danielle Albers Szafr. 2022. Making data tangible: A cross-disciplinary design space for data physicalization. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.
- Catherine M. Baker, Yasmine N. El-Glaly, and Kristen Shinohara. 2020. *A Systematic Analysis of Accessibility in Computing Education Research*. Association for Computing Machinery, New York, NY, USA, 107–113. <https://doi.org/10.1145/3328778.3366843>
- Marta G. Carcedo, Soon Hau Chua, Simon Perrault, Pawel Wozniak, Raj Joshi, Mohammad Obaid, Morten Fjeld, and Shengdong Zhao. 2016. HapticColor: Interpolating Color Information as Haptic Feedback to Assist the Colorblind. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 3572–3583. <https://doi.org/10.1145/2858036.2858220>
- Michael E. Caspersen and Henrik Bærak Christensen. 2000. Here, There and Everywhere - on the Recurring Use of Turtle Graphics in CS1. In *Proceedings of the Australasian Conference on Computing Education* (Melbourne, Australia) (ACSE '00). Association for Computing Machinery, New York, NY, USA, 34–40. <https://doi.org/10.1145/359369.359375>